

Math Bootcamp

Day 2 — More calculus and existence

Math Camp 2026

How to use these slides

Main deck: definitions, recipes, examples, and practice.

Appendix and proofs (marked on relevant slides): **for interest only** — optional recap for those already comfortable with the math.

Not required for Math Camp. Focus on the main slides and exercises unless you want the extra detail.

Don't make your partner disappointed

 新浪网

 司天钦Gosta
26-02-03 11:50 微博认证: 美食博主

女生男生都超级扣分的行为，快来看看自己中了几条

1. 不定积分结果不加常数C
2. 定积分换元时忘记更换积分上下限
3. 滥用洛必达法则
4. 求复合函数导数时遗漏链式法则项
5. 判别级数敛散性误用判别法
6. 计算行列式或逆矩，目属代数余子式的符号
7. 混淆偏导数与全导数
8. 分部积分法中 du 和 dv 选择不当
9. 在加减运算中，误用等价无穷小
10. 解非齐次微分方程时遗漏特解

2.1 Endogenous vs. exogenous — supply and demand

In a competitive market, demand and supply determine an equilibrium **price** and **quantity**. Not everything in that picture is determined the same way.

Exogenous (taken as given, from **outside** the model): things that shift the curves — harvest / weather, costs, income, tastes, a tax, population, ... Call a typical shifter θ .

Endogenous (solved for **inside** the model): equilibrium price p^* and quantity q^* . They adjust until $D = S$.

Slopes of D and S are also usually treated as exogenous parameters. The model takes them as given and then finds p^*, q^* .

2.1 Example — eggs in a competitive market

Market. Many buyers and many sellers of eggs; take the market price as given (competitive). Demand is for eggs as a daily staple.

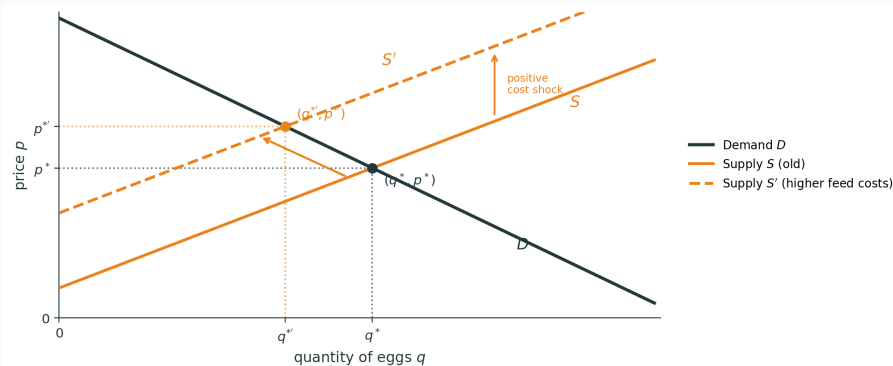
Exogenous (shifters, not chosen in this market):

- feed costs, weather / disease on farms (supply)
- household income, taste for breakfast, a tax or subsidy (demand or supply)

Endogenous (solved in this market): the egg price p^* and quantity sold q^* .

If feed costs jump (exogenous supply shock), the supply curve shifts. Then endogenous p^* and q^* adjust to a new equilibrium.

2.1 Egg market — a feed-cost shock



This figure: **positive** cost shock = **negative** supply shock (S left; p^* up, q^* down). Converse: **negative** cost shock = **positive** supply shock (S right).

2.1 Motivation — comparative statics

Once p^* is determined by exogenous θ , a natural question is: if θ changes, how does the **endogenous** equilibrium move?

Comparative statics: how $p^*(\theta)$ (and $q^*(\theta)$) respond to a shift in an exogenous parameter.
We want $\frac{dp^*}{d\theta}$.

Eggs. Higher feed costs $\Rightarrow$ supply shifts left $\Rightarrow p^*$ typically rises and q^* falls. We want a systematic way to get the sign and magnitude.

Same pattern for a demand shifter (income, tastes), a tax, ... The tool is the same once equilibrium is written as $F(p, \theta) = 0$.

2.1 Market clearing

Quantity demanded equals quantity supplied:

$$D(p, \theta) = S(p, \theta).$$

Exogenous θ is a shifter (supply shock, demand shock, tax, ...); endogenous p and q adjust.

Excess demand

$$E(p, \theta) = D(p, \theta) - S(p, \theta)$$

is zero at equilibrium: $E(p^*(\theta), \theta) = 0$.

Question. How do p^* and q^* change when θ changes?

2.1 Linear market — an analytical solution

Linear demand and supply; supply shifter s , demand intercept α :

$$D(p, \alpha) = \alpha - bp, \quad S(p, s) = s + dp, \quad b, d > 0.$$

Market clearing $D = S$ gives both endogenous variables in closed form:

$$p^*(s, \alpha) = \frac{\alpha - s}{b + d}, \quad q^*(s, \alpha) = \alpha - bp^* = \frac{\alpha d + bs}{b + d}.$$

We can differentiate these formulas directly — no new theorem yet.

2.1 Supply shock vs. demand shock

Positive supply shock ($s \uparrow$; equivalently a negative cost shock):

$$\frac{dp^*}{ds} = -\frac{1}{b+d} < 0, \quad \frac{dq^*}{ds} = \frac{b}{b+d} > 0.$$

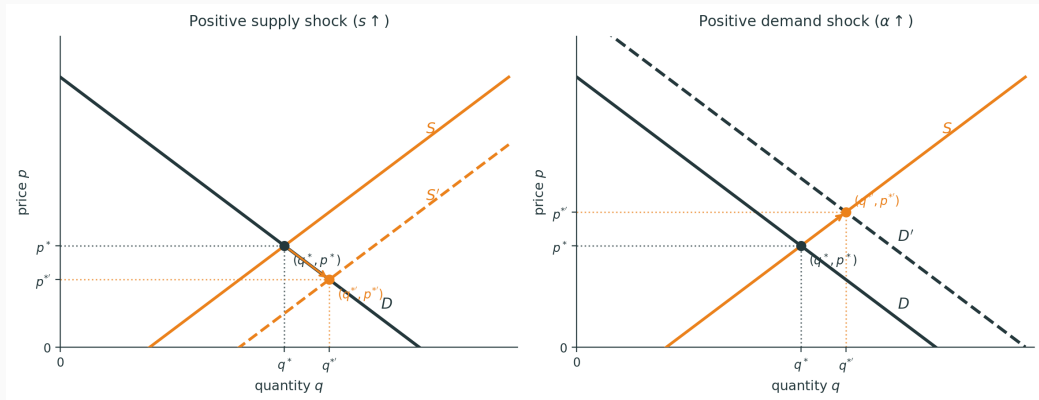
Price falls, quantity rises: p^* and q^* move in **opposite** directions.

Positive demand shock ($\alpha \uparrow$; income, tastes, ...):

$$\frac{dp^*}{d\alpha} = \frac{1}{b+d} > 0, \quad \frac{dq^*}{d\alpha} = \frac{d}{b+d} > 0.$$

Both price and quantity rise: p^* and q^* move in the **same** direction.

2.1 Supply shock vs. demand shock



Left: $s \uparrow$ — p^* down, q^* up. Right: $\alpha \uparrow$ — both p^* and q^* up.

2.1 Explicit vs. implicit

Explicit (what we just did): we solved for p^* and q^* as formulas in the exogenous variables, e.g.

$$p^* = g(s, \alpha) = \frac{\alpha - s}{b + d}.$$

Then ordinary differentiation gives $\frac{dp^*}{ds}$, $\frac{dq^*}{ds}$, and the demand-shock derivatives.

Implicit: we only have an equation $F(p, \theta) = 0$ (e.g. $E(p, \theta) = 0$). We **know** F , but may not be able to write $p^* = g(\theta)$ in closed form.

Next: a nonlinear market where we cannot solve for p^* .

2.1 No closed form — nonlinear market

Now change demand to

$$D(p, \alpha) = \alpha e^{-p}, \quad S(p) = p, \quad \alpha > 0.$$

Clearing: $\alpha e^{-p} = p$, or $E(p, \alpha) = \alpha e^{-p} - p = 0$.

This equation **defines** $p^*(\alpha)$, but it has **no elementary closed form**. We cannot write $p^* = g(\alpha)$ and differentiate.

We still want $\frac{dp^*}{d\alpha}$. That is the problem the Implicit Function Theorem solves.

2.1 What problem does the IFT solve?

An equilibrium condition $F(x, \theta) = 0$ defines $x^*(\theta)$ **implicitly**. We want $\frac{dx^*}{d\theta}$ without solving for $x^*(\theta)$.

IFT: if F is well-behaved and $F_x \neq 0$ at the equilibrium, then $x^*(\theta)$ exists locally as a function, and

$$\frac{dx^*}{d\theta} = -\frac{F_\theta}{F_x}$$

— computed from partials of F only.

Same idea as the linear market, but it still works when there is no formula for $x^*(\theta)$.

2.1 One-variable IFT — statement and recipe

Let $F(x, y)$ be C^1 near (x_0, y_0) with $F(x_0, y_0) = 0$. If $F_y(x_0, y_0) \neq 0$, then locally there is a unique g such that

$$y = g(x), \quad F(x, g(x)) = 0, \quad g(x_0) = y_0,$$

and

$$\boxed{g'(x) = \frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)}} \quad \text{on the curve } F(x, y) = 0.$$

Recipe. Totally differentiate $F(x, y) = 0$:

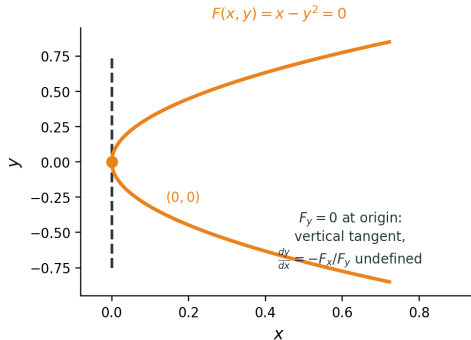
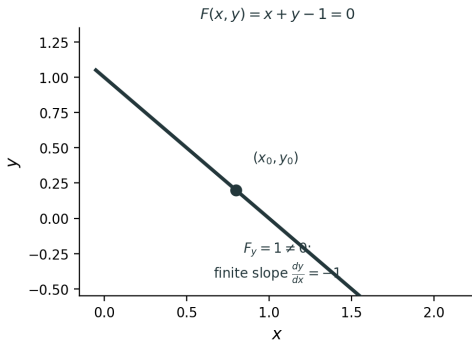
$$0 = dF = F_x dx + F_y dy \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{F_x}{F_y} \quad \text{when } F_y \neq 0.$$

Several variables (**for interest only**): [appendix](#).

2.1 Why $F_y \neq 0$? — regularity condition

Left: $F_y \neq 0$ at (x_0, y_0) — tangent has finite slope; locally $y = g(x)$. **Right:** $F_y = 0$ at $(0, 0)$ — **vertical tangent**; $\frac{dy}{dx} = -F_x/F_y$ is undefined.

Why the IFT needs $F_y \neq 0$



2.1 Check — IFT recovers the linear answer

Return to $E(p, s) = \alpha - s - (b + d)p$. Partial: $E_p = -(b + d) \neq 0$, $E_s = -1$.

IFT:

$$\frac{dp^*}{ds} = -\frac{E_s}{E_p} = -\frac{-1}{-(b+d)} = -\frac{1}{b+d} < 0.$$

Same as differentiating $p^*(s) = (\alpha - s)/(b + d)$ explicitly.

Interpretation (direct vs. indirect). Higher s lowers excess demand at the old price ($E_s < 0$: surplus).

Price must fall ($E_p < 0$) until $E = 0$ is restored.

2.1 — Apply IFT (no closed form) Together

Nonlinear demand: $D(p, \alpha) = \alpha e^{-p}$, $S(p) = p$, $\alpha > 0$.

No elementary closed form for $p^*(\alpha)$. Use IFT.

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2. Partial derivatives E_α , E_p .

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4. On the curve $\alpha e^{-p^*} = p^*$, simplify: $\frac{dp^*}{d\alpha} = \frac{e^{-p^*}}{p^* + 1} > 0$

5. $\alpha = 2$: $2e^{-p} = p$ gives $p^* \approx 0.85$, so $\frac{dp^*}{d\alpha} \approx 0.23$.

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No closed form for $p^*(\alpha)$ — IFT still works.

2.1 Example — tax incidence

Setup. Demand $D(p) = \alpha - bp$, supply $S(p) = c + dp$ ($b, d > 0$). A per-unit tax t on **sellers**: buyers pay p , sellers receive $p - t$. (A tax is another supply-side shock.)

Clearing: $D(p) = S(p - t)$, so $E(p, t) = \alpha - bp - c - d(p - t) = 0$.

Partials: $E_p = -(b + d)$, $E_t = d$. **IFT:**

$$\frac{dp^*}{dt} = -\frac{E_t}{E_p} = \frac{d}{b + d} > 0.$$

Incidence (per \$1 of tax). Buyer price rises by $\frac{d}{b + d}$; seller net price falls by $\frac{b}{b + d}$. The tax is borne **more heavily by the less elastic side** (small b : inelastic demand $\Rightarrow$ buyers pay most; small d : inelastic supply $\Rightarrow$ sellers pay most).

2.2 EVT on a closed interval $[a, b]$

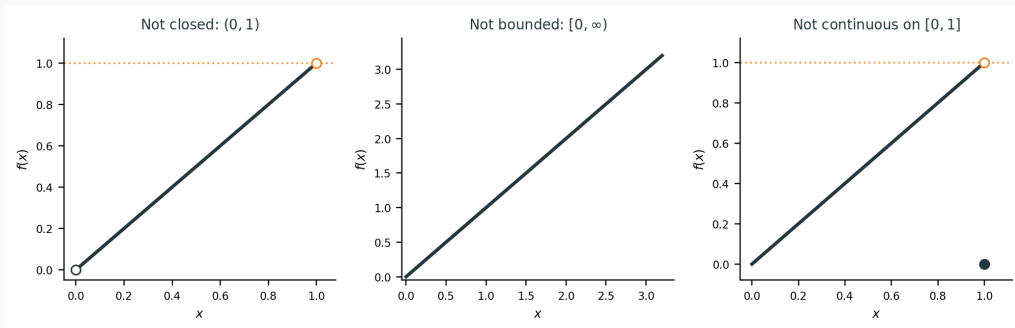
Extreme Value Theorem (EVT). If f is **continuous** on the **closed** interval $[a, b]$, then f attains a **maximum** and a **minimum** on $[a, b]$.

So $\exists c, d \in [a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \quad \text{for all } x \in [a, b].$$

$[a, b]$ is **closed and bounded** — a compact set in $\mathbb{R}$.

2.2 When EVT fails



Need continuity on a **closed, bounded** domain (or compact set more generally).

2.2 Open and closed sets

Open set (intuition): every point has a small neighborhood still inside the set.

Closed set (intuition): contains its **boundary**; limits of sequences in the set stay in the set.

Economic examples.

- Budget $\{x : p \cdot x \leq B\}$ — **closed** (includes the budget line).
- Strict budget $p \cdot x < B$ — **not closed** (boundary excluded).
- Reservation price set $P = \{p : p < \bar{p}\}$ — not closed; $\sup P = \bar{p}$ may not be attained.

On $\mathbb{R}^1$: $[a, b]$ is closed and bounded; (a, b) is open (not closed). The failure figure above: not closed, not bounded, or not continuous.

2.2 Compact sets and Weierstrass EVT

In $\mathbb{R}^n$ (camp level): a set is **compact** when it is **closed and bounded** (Heine–Borel, stated).

Weierstrass Extreme Value Theorem. If f is continuous on a compact set $K \subset \mathbb{R}^n$, then f attains a **maximum** and **minimum** on K .

Why economists care.

compact feasible set K + continuous objective $f \Rightarrow$ optimum **exists**

(preview of Day 4 optimization).

Proof: **appendix (for interest only)**.

2.2 Example — existence of a utility maximizer

Setup (consumer problem). Prices $p_1, p_2 > 0$, income $w > 0$, budget set

$$\mathcal{B} = \{(x_1, x_2) \in \mathbb{R}_+^2 : p_1 x_1 + p_2 x_2 \leq w\}.$$

1. $\mathcal{B}$ is **closed** (weak inequality $\leq$) and **bounded** (e.g. $0 \leq x_i \leq w/p_i$), hence **compact** in $\mathbb{R}^2$.
2. Suppose utility $u(x_1, x_2)$ is **continuous** on $\mathcal{B}$.
3. By Weierstrass EVT, u attains a **maximum** on $\mathcal{B}$.

Conclusion. An optimal consumption bundle $x^* \in \mathcal{B}$ **exists** (existence theorem for the consumer problem; first-order conditions come on Day 4).

Not uniqueness. Weierstrass gives at least one maximizer, not a unique one (e.g. $u(x_1, x_2) = x_1 + x_2$ has many optima on $\mathcal{B}$). Uniqueness needs extra conditions (e.g. strictly concave / strictly quasiconcave u ; Day 4).

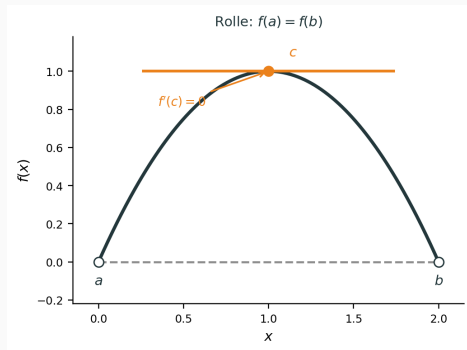
2.2 Rolle's theorem

Rolle's theorem. If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ with $f'(c) = 0$.

for interest only

Proof. If f is constant on $[a, b]$, any $c \in (a, b)$ works. Otherwise, by EVT, f has a maximum or minimum on $[a, b]$. Since $f(a) = f(b)$, this extremum occurs at some **interior** point $c \in (a, b)$. At an interior peak or trough, the tangent is horizontal, so $f'(c) = 0$. $\square$

Special case of the MVT below with $f(a) = f(b)$.



2.2 Mean Value Theorem

Mean Value Theorem (MVT). If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that

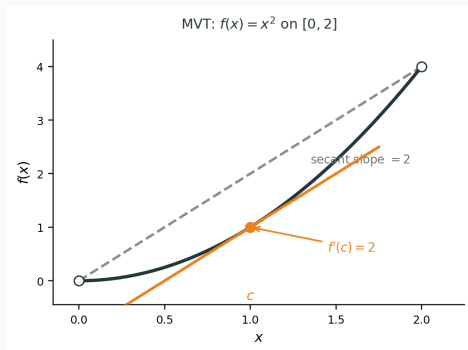
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Picture: some interior tangent is **parallel** to the secant through $(a, f(a))$ and $(b, f(b))$.

Example. $f(x) = x^2$ on $[0, 2]$: secant slope = 2, so $f'(c) = 2c = 2$ gives $c = 1$.

for interest only

Proof idea: define $g(x) = f(x) - m(x - a)$ with $m = \frac{f(b) - f(a)}{b - a}$; then $g(a) = g(b)$, Rolle gives $f'(c) = m$.



2.2 Example — labor and output (economics)

Setup. Output $f(L)$ from labor L ; f continuous on $[L_1, L_2]$, differentiable on (L_1, L_2) .

MVT. $\exists L^* \in (L_1, L_2)$ with

$$f'(L^*) = \underbrace{\frac{f(L_2) - f(L_1)}{L_2 - L_1}}_{\text{average product gain over } [L_1, L_2]} .$$

At some interior employment level, **marginal product** = **average product** of the change.

2.2 MC and AC — what the crossover means

Cost example. $C(q) = q^2 + 4$ (fixed cost 4, rising marginal cost).

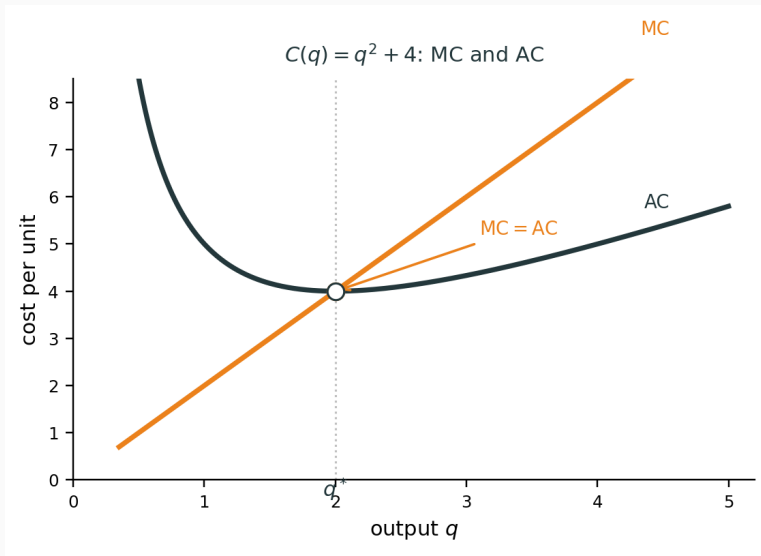
$$MC(q) = C'(q) = 2q, \quad AC(q) = \frac{C(q)}{q} = q + \frac{4}{q}.$$

MVT link. On $[0, \bar{q}]$, $\exists q^* \in (0, \bar{q})$ with $MC(q^*) = \frac{C(\bar{q}) - C(0)}{\bar{q}} = \bar{q}$: marginal cost somewhere equals the **average incremental cost** of the first $\bar{q}$ units. (If $C(0) = 0$, this is $AC(\bar{q})$.)

Why MC crosses AC at the minimum of AC. Minimizing $AC(q) = C(q)/q$ gives $MC(q) = C(q)/q$ (FOC) $\Rightarrow$ curves cross at $q^* = 2$.

- For $q < q^*$: $MC < AC \Rightarrow$ an extra unit pulls the average **down**.
- For $q > q^*$: $MC > AC \Rightarrow$ an extra unit pulls the average **up**.
- At q^* : the last unit costs exactly the average — **efficient scale** (lowest cost per unit).

2.2 MC and AC — graph



2.3 Integral as area and accumulation

Riemann idea. For $f \geq 0$ on $[a, b]$,

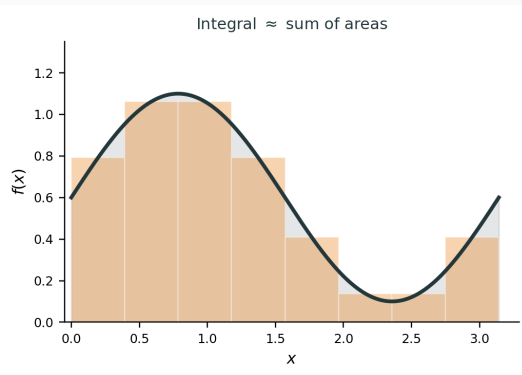
$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i^*) \Delta x$$

is the **signed area** under the curve / total accumulation.

Definite vs indefinite.

- Definite: $\int_a^b f(x) dx$ is a number
- Indefinite: $\int f(x) dx = F(x) + C$ is a family of antiderivatives

Econ: total cost, consumer surplus, probability mass.



2.3 Riemann vs. Lebesgue — for interest

Not required for Math Camp.

Two ways to define $\int f$.

Riemann (this camp)

Chop the x -axis into small pieces

Sum (height) $\times$ (width)

Good for hand calculations

Area under curve picture

Lebesgue (grad / probability)

Chop the y -axis into levels

Sum (level) $\times$ (measure of set where f hits that level)

Integrates more pathological functions

Foundation for modern probability

Contrast in one sentence. Riemann slices **vertically**; Lebesgue slices **horizontally** by value.

2.3 Lebesgue integral — for interest

Not required for Math Camp.

When Riemann fails. Riemann needs the function to be “well behaved” on intervals: too many jumps can prevent upper and lower sums from converging to the same number.

Classic example (not Riemann integrable). Dirichlet function on $[0, 1]$:

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Every interval contains rationals and irrationals, so upper and lower Riemann sums never agree — no Riemann integral.

Lebesgue handles this. It integrates by **level sets**, so a “sparse” jump set like $\mathbb{Q}$ does not block integration the way Riemann’s interval partitions do.

2.3 Fundamental Theorem of Calculus

Part I. If $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Part II. If f is continuous and $F(x) = \int_a^x f(t) dt$, then $F'(x) = f(x)$.

Differentiation and integration are inverse operations (MVT underlies many proofs).

2.3 Integration by parts

From the product rule. For differentiable $u(x)$ and $v(x)$,

$$(uv)' = u'v + uv' \quad \Rightarrow \quad uv' = (uv)' - u'v.$$

Integrate both sides:

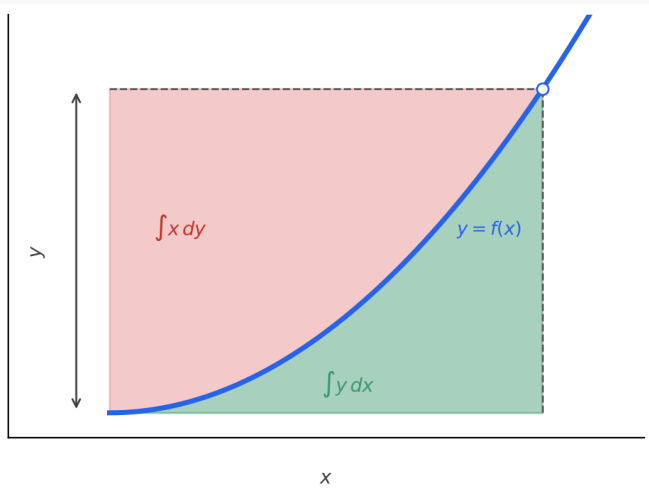
$$\int u v' dx = uv - \int u' v dx \quad \text{or} \quad \boxed{\int u dv = uv - \int v du}.$$

On $[a, b]$ with $u'v$ integrable,

$$\int_a^b u(x)v'(x) dx = \left[u(x)v(x) \right]_a^b - \int_a^b u'(x)v(x) dx.$$

Pick u and dv so the new integral $\int v du$ is simpler.

2.3 Integration by parts (graph)



2.3 Integration by parts — example

Example. $\int x e^x dx$.

Let $u = x$, $dv = e^x dx$, so $du = dx$, $v = e^x$:

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C = e^x(x - 1) + C.$$

Check. $\frac{d}{dx} [e^x(x - 1)] = e^x(x - 1) + e^x = x e^x$. ✓

2.3 PMF and CMF — discrete example (dice)

Setup. Fair die: $X \in \{1, 2, 3, 4, 5, 6\}$.

Probability mass function (PMF): $p(k) = P(X = k) = \frac{1}{6}$ for $k = 1, \dots, 6$.

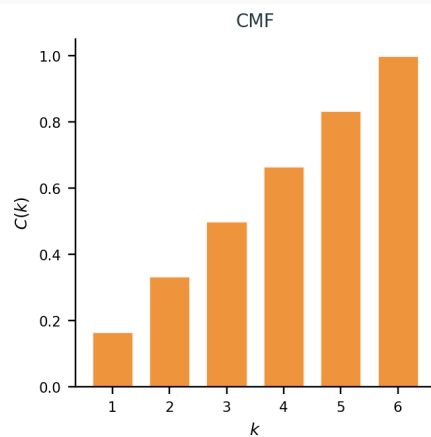
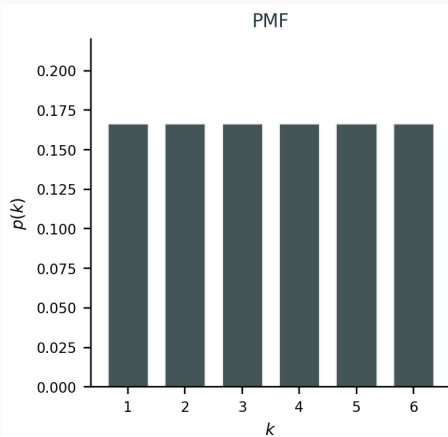
Event (sum rule): $P(2 \leq X \leq 4) = p(2) + p(3) + p(4) = \frac{1}{2}$.

Cumulative mass function (CMF): $C(k) = P(X \leq k) = \sum_{j \leq k} p(j)$, defined only at the **outcomes** $k = 1, \dots, 6$ (not a continuous curve in x).

Example: $C(3) = p(1) + p(2) + p(3) = \frac{1}{2}$.

Discrete: **PMF/CMF** and **sums**. Continuous (next slides): **PDF/CDF** and **integrals**.

2.3 PMF and CMF — dice (graph)



2.3 PDF and CDF — continuous case

Let X be a continuous random variable.

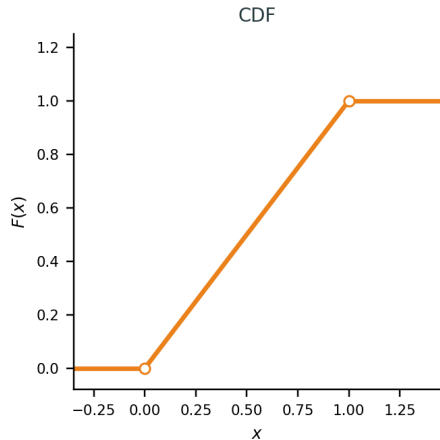
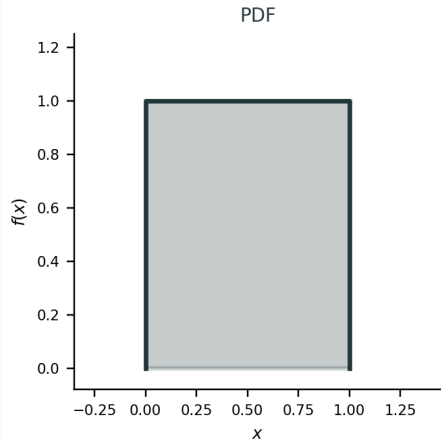
Probability density function (PDF): $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x) dx = 1$. Then

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

Cumulative distribution function (CDF): $F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$; a **continuous** curve in x (not jumps on a finite outcome list).

FTC: if f is continuous, then $F'(x) = f(x)$.

2.3 PDF and CDF — Unif(0, 1) (graph)



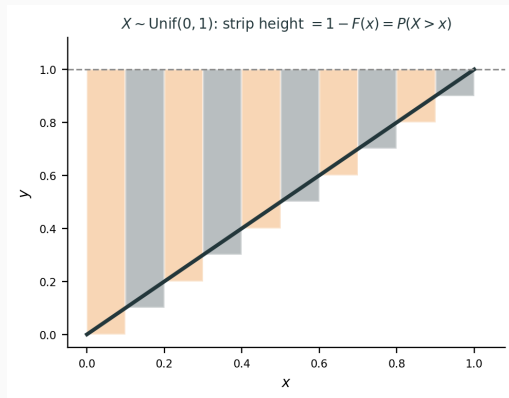
2.3 Expectation from the CDF

For a **non-negative** continuous X with PDF f and CDF F ,

$$\mathbb{E}[X] = \int_0^{\infty} xf(x) dx = \int_0^{\infty} (1 - F(x)) dx.$$

Intuition. For $X \sim \text{Unif}(0, 1)$, each vertical strip has height $1 - F(x)$; $\mathbb{E}[X]$ is their total area.

Derivation via integration by parts (**for interest only**):
[appendix](#).



2.4 Leibniz rule — fixed limits

Setup. $F(\theta) = \int_a^b f(x, \theta) dx$ with fixed a, b .

Leibniz rule (fixed limits). When regularity holds,

$$\frac{dF}{d\theta} = \int_a^b \frac{\partial f}{\partial \theta}(x, \theta) dx.$$

Regularity (enough for this camp).

- $\frac{\partial f}{\partial \theta}(x, \theta)$ exists
- f and $\frac{\partial f}{\partial \theta}$ are **continuous** in (x, θ)

Then you may swap $\frac{d}{d\theta}$ and $\int_a^b$.

Idea. Only the integrand moves with θ ; the limits a, b stay fixed.

When regularity fails (**for interest only**): [appendix](#).

2.4 Leibniz rule — variable limits

Setup. $F(\theta) = \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx.$

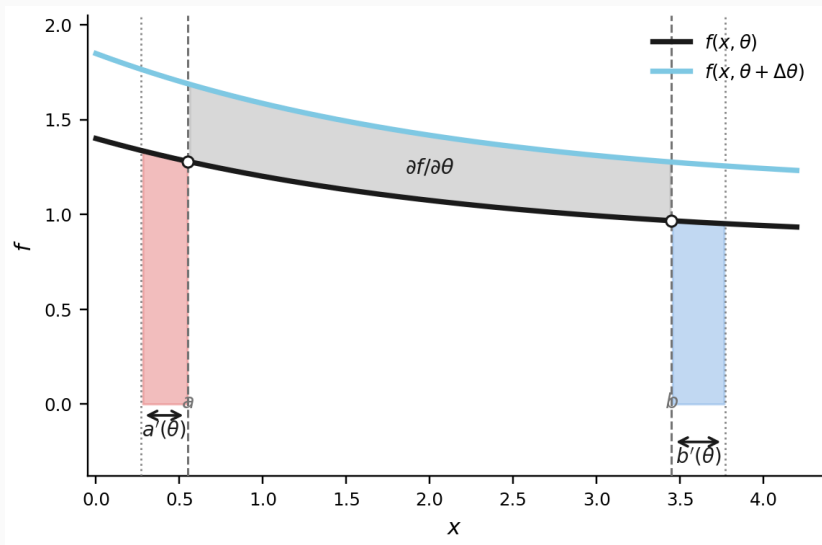
Leibniz rule (variable limits). When $a(\theta)$, $b(\theta)$ are differentiable and regularity holds,

$$\frac{dF}{d\theta} = f(b(\theta), \theta) b'(\theta) - f(a(\theta), \theta) a'(\theta) + \int_{a(\theta)}^{b(\theta)} \frac{\partial f}{\partial \theta}(x, \theta) dx.$$

Three parts. (i) area added/lost at the **upper** limit b ; (ii) at the **lower** limit a ; (iii) change **inside** the interval $(\partial f / \partial \theta)$.

Graph on next slide: grey = $\partial f / \partial \theta$; pink/blue = lower/upper limit terms $(a'(\theta), b'(\theta))$.

2.4 Leibniz rule — variable limits (graph)



2.4 Leibniz — practice (monopoly pricing)

Economic setup. A unit mass of consumers; consumer i has willingness to pay v_i , with $v_i \sim F$ (cdf F , pdf f on $[0, \bar{v}]$). A monopolist sets price p ; i buys iff $v_i \geq p$.

Type-dependent cost. Serving type v costs $c(v)$ (e.g. quality, personalization, delivery). Profit from a buyer with valuation $v \geq p$ is $p - c(v)$ — margin **varies by type**.

Total profit (no simple closed form).

$$\pi(p) = \int_p^{\bar{v}} (p - c(v)) f(v) dv \quad (\text{cannot write as } (p - c)(1 - F(p)) \text{ when } c = c(v)).$$

The lower limit moves with p — **Leibniz is the right tool**.

Practice. Find $\frac{d\pi}{dp}$ and interpret each term. If c is constant, product rule also works; here it does not.

2.4 Leibniz — practice solution (monopoly pricing)

Apply variable-limits Leibniz to $\pi(p)$ directly. Lower limit $a(p) = p$, upper limit $\bar{v}$; only p in the integrand is the leading p :

$$\frac{d\pi}{dp} = -(p - c(p))f(p) a'(p) + \int_p^{\bar{v}} \frac{\partial}{\partial p} [(p - c(v))f(v)] dv.$$

Simplify. $a'(p) = 1$; $\frac{\partial}{\partial p} [(p - c(v))f(v)] = f(v)$ (since $c(v)$ does not depend on p):

$$\frac{d\pi}{dp} = -(p - c(p))f(p) + \int_p^{\bar{v}} f(v) dv = -(p - c(p))f(p) + (1 - F(p)).$$

Economic reading.

- **Lower limit** $-(p - c(p))f(p)$: lose margin $p - c(p)$ on the marginal buyer ($v = p$).
- **Inside** $1 - F(p)$: raising p still adds $1 \cdot dp$ profit on **each** remaining buyer.

FOC. $(p - c(p))f(p) = 1 - F(p)$; constant c gives the usual markup rule $p - c = \frac{1 - F(p)}{f(p)}$.

2.4 Fubini's theorem

Statement. If f is **integrable** on region D , i.e.

$$\iint_D |f(x, y)| \, dx \, dy < \infty,$$

then

$$\iint_D f(x, y) \, dx \, dy = \int \left(\int f(x, y) \, dx \right) dy = \int \left(\int f(x, y) \, dy \right) dx,$$

(inner limits depend on the shape of D).

Camp shortcut. If D is bounded and f is **continuous** on D , then f is integrable and Fubini applies.

Tonelli (for interest): if $f \geq 0$ and measurable, the same swap holds without checking $\iint |f|$ first.

2.4 Fubini — triangle example

Example (triangle).

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}.$$

Compute $\iint_D (x + y) \, dx \, dy$.

y first, then x: for fixed x , $y \in [0, x]$:

$$\int_0^1 \int_0^x (x + y) \, dy \, dx = \int_0^1 \left[xy + \frac{y^2}{2} \right]_0^x dx = \int_0^1 \frac{3x^2}{2} \, dx = \frac{1}{2}.$$

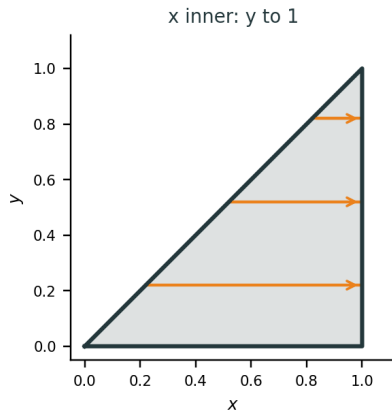
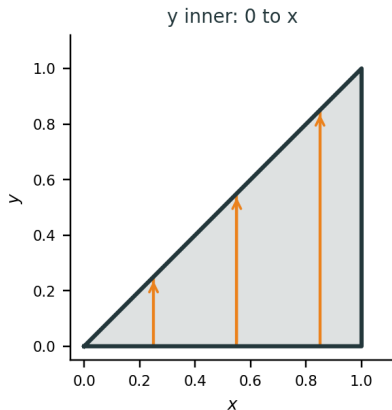
x first, then y: for fixed y , $x \in [y, 1]$:

$$\int_0^1 \int_y^1 (x + y) \, dx \, dy = \int_0^1 \left[\frac{x^2}{2} + xy \right]_y^1 dy = \int_0^1 \left(\frac{1}{2} + y - \frac{3y^2}{2} \right) dy = \frac{1}{2}.$$

Graph on next slide: same triangle, different slice directions.

2.4 Fubini — triangle (graph)

Triangle D : $0 \leq x \leq 1$, $0 \leq y \leq x$



2.4 Fubini — practice (between two curves)

Region (non-rectangular). Between the parabola $y = x^2$ and the line $y = x$ on $[0, 1]$:

$$D = \{(x, y) : 0 \leq x \leq 1, x^2 \leq y \leq x\}.$$

(On $(0, 1)$, the line lies **above** the parabola.)

Practice. Compute $\iint_D xy \, dx \, dy$ using Fubini **twice** (different inner variable).

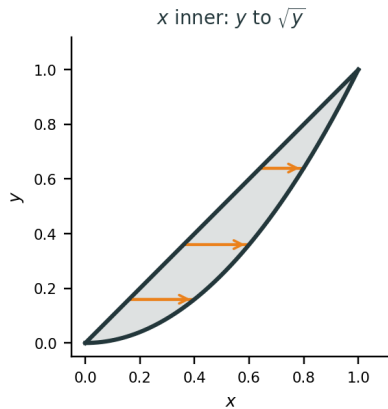
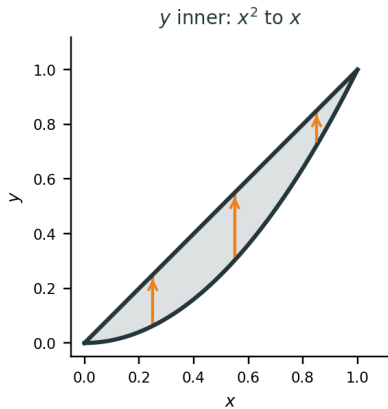
Slice limits to find.

- **y inner:** for fixed $x \in [0, 1]$, y runs from x^2 to x .
- **x inner:** for fixed $y \in [0, 1]$, solve $x^2 \leq y \leq x$ to get $x \in [y, \sqrt{y}]$.

Graph on next slide; detailed solution on the following slides.

2.4 Fubini — practice (graph)

Region D : $x^2 \leq y \leq x$ on $0 \leq x \leq 1$



2.4 Fubini — practice solution (y inner first)

Order 1: $\int_0^1 \int_{x^2}^x xy \, dy \, dx.$

Inner integral (fix x).

$$\int_{x^2}^x xy \, dy = x \left[\frac{y^2}{2} \right]_{y=x^2}^{y=x} = x \left(\frac{x^2}{2} - \frac{x^4}{2} \right) = \frac{x^3}{2} - \frac{x^5}{2}.$$

Outer integral.

$$\int_0^1 \left(\frac{x^3}{2} - \frac{x^5}{2} \right) dx = \left[\frac{x^4}{8} - \frac{x^6}{12} \right]_0^1 = \frac{1}{8} - \frac{1}{12} = \frac{1}{24}.$$

2.4 Fubini — practice solution (x inner first)

Order 2: $\int_0^1 \int_y^{\sqrt{y}} xy \, dx \, dy.$

Why $x \in [y, \sqrt{y}]$? For fixed y , need $y \leq x$ (from $y \leq x$) and $x^2 \leq y$ (from $y \geq x^2$), i.e. $x \leq \sqrt{y}$.

Inner integral (fix y).

$$\int_y^{\sqrt{y}} xy \, dx = y \left[\frac{x^2}{2} \right]_{x=y}^{x=\sqrt{y}} = y \left(\frac{y}{2} - \frac{y^2}{2} \right) = \frac{y^2}{2} - \frac{y^3}{2}.$$

Outer integral.

$$\int_0^1 \left(\frac{y^2}{2} - \frac{y^3}{2} \right) dy = \left[\frac{y^3}{6} - \frac{y^4}{8} \right]_0^1 = \frac{1}{6} - \frac{1}{8} = \frac{1}{24}.$$

Check. Both orders agree — Fubini is consistent when the region and integrand are well behaved.

2.5 Taylor expansion — formal statement

One variable. If f has n derivatives at a ,

$$f(a + h) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} h^k + R_n(h), \quad R_n(h) \rightarrow 0 \text{ faster than } h^n.$$

Several variables ($f: \mathbb{R}^m \rightarrow \mathbb{R}$, $\mathbf{a} \in \mathbb{R}^m$). If f is C^2 near $\mathbf{a}$,

$$f(\mathbf{a} + \mathbf{h}) = f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h} + o(\|\mathbf{h}\|^2),$$

$H_f(\mathbf{a})$ = Hessian at $\mathbf{a}$ (matrix of second partials).

Proof (one variable): [appendix \(for interest only\)](#).

2.5 First-order approximation

One variable (truncate at $k = 1$):

$$f(a + h) \approx f(a) + f'(a) h.$$

Several variables:

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h}.$$

2.5 Second-order approximation

One variable (truncate at $k = 2$):

$$f(a + h) \approx f(a) + f'(a) h + \frac{1}{2} f''(a) h^2.$$

Several variables:

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h}.$$

Use. Captures **curvature** (concavity, cross-partials); improves forecasts when the first-order term is not enough.

Itô's lemma: [appendix](#) (for interest only).

2.5 — First-order Taylor practice (two variables) Together

Approximate $f(1.1, 1.2)$ for $f(x, y) = x^2 + y$ using Taylor at $(1, 1)$, $\mathbf{h} = (0.1, 0.2)$.

2.5 — First-order Taylor practice (two variables) Together

Approximate $f(1.1, 1.2)$ for $f(x, y) = x^2 + y$ using Taylor at $(1, 1)$, $\mathbf{h} = (0.1, 0.2)$.

At $(1, 1)$: $f(1, 1) = 2$, $f_x = 2$, $f_y = 1$.

2.5 — First-order Taylor practice (two variables) Together

Approximate $f(1.1, 1.2)$ for $f(x, y) = x^2 + y$ using Taylor at $(1, 1)$, $\mathbf{h} = (0.1, 0.2)$.

At $(1, 1)$: $f(1, 1) = 2$, $f_x = 2$, $f_y = 1$.

First-order: $2 + 2(0.1) + 1(0.2) = 2.4$.

2.5 — First-order Taylor practice (two variables) Together

Approximate $f(1.1, 1.2)$ for $f(x, y) = x^2 + y$ using Taylor at $(1, 1)$, $\mathbf{h} = (0.1, 0.2)$.

At $(1, 1)$: $f(1, 1) = 2$, $f_x = 2$, $f_y = 1$.

First-order: $2 + 2(0.1) + 1(0.2) = 2.4$.

Exact $f(1.1, 1.2) = 2.41$.

2.5 When linearization works

Works well when:

- the shift is **small**
- curvature is modest (f'' not huge)
- the function is differentiable

Fails or misleads when:

- large shifts, kinks, or non-differentiability
- strong nonlinearity (e.g. corner solutions, thresholds)

2.5 Two variables — matrix form unpacked ignore if you are not familiar with what's Hessian

Setup. $\mathbf{a} = (a_1, a_2)$, $\mathbf{h} = (h_1, h_2)$:

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + \nabla f(\mathbf{a}) \cdot \mathbf{h} + \frac{1}{2} \mathbf{h}^T H_f(\mathbf{a}) \mathbf{h}.$$

First-order. $\nabla f = (f_{x_1}, f_{x_2})$ at $\mathbf{a}$, so $\nabla f \cdot \mathbf{h} = f_{x_1} h_1 + f_{x_2} h_2$.

Second-order. $H_f = \begin{pmatrix} f_{x_1 x_1} & f_{x_1 x_2} \\ f_{x_2 x_1} & f_{x_2 x_2} \end{pmatrix}$, and multiply:

$$\mathbf{h}^T H_f \mathbf{h} = (h_1 \ h_2) \begin{pmatrix} f_{x_1 x_1} h_1 + f_{x_1 x_2} h_2 \\ f_{x_2 x_1} h_1 + f_{x_2 x_2} h_2 \end{pmatrix} = f_{x_1 x_1} h_1^2 + (f_{x_1 x_2} + f_{x_2 x_1}) h_1 h_2 + f_{x_2 x_2} h_2^2.$$

If $f \in C^2$, then $f_{x_1 x_2} = f_{x_2 x_1}$, hence

$$f(\mathbf{a} + \mathbf{h}) \approx f(\mathbf{a}) + f_{x_1} h_1 + f_{x_2} h_2 + \frac{1}{2} (f_{x_1 x_1} h_1^2 + 2f_{x_1 x_2} h_1 h_2 + f_{x_2 x_2} h_2^2).$$

2.5 — Second-order Taylor practice (two variables)

Take home

ignore if you are

not familiar with what's Hessian

Approximate $f(0.1, 0.2)$ for $f(x, y) = \sin x + \cos y$ using a second-order Taylor expansion at $(0, 0)$.

Use $\mathbf{h} = (0.1, 0.2)$ and the explicit second-order formula on the matrix-form slide.

Check your answer with a calculator (no need to simplify trig decimals).

2.6 Pointwise vs. uniform convergence

Pointwise. $f_n \rightarrow f$ on D means: for each fixed $x \in D$,

$$f_n(x) \rightarrow f(x) \quad \text{as } n \rightarrow \infty.$$

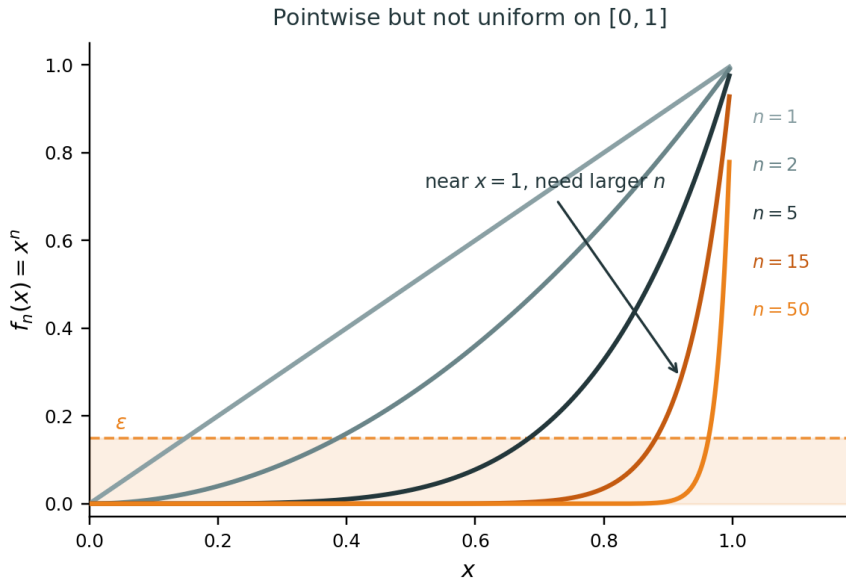
Equivalently: for each x and every $\varepsilon > 0$, $\exists N$ (may depend on **both** x **and** ε) such that $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$.

Uniform. For every $\varepsilon > 0$, $\exists N$ (depends on ε **only**) such that for all $n \geq N$ and **all** $x \in D$,

$$|f_n(x) - f(x)| < \varepsilon.$$

One N works **uniformly** over the whole domain.

2.6 Example — pointwise but not uniform



2.6 Why uniform convergence matters

Uniform convergence is the safe condition for swapping limits with other operations:

- **Continuity:** if each f_n is continuous and $f_n \rightarrow f$ **uniformly**, then f is continuous.
- **Integration:** under uniform convergence, $\lim_{n \rightarrow \infty} \int f_n = \int \lim_{n \rightarrow \infty} f_n$.
- **Differentiation** (with extra hypotheses): uniform convergence of derivatives can justify $\frac{d}{dx} \lim f_n = \lim \frac{d}{dx} f_n$.
- **Example** (x^n on $[0, 1]$): each f_n is continuous, but the pointwise limit (0 on $[0, 1)$, 1 at $x = 1$) jumps at $x = 1$ — **continuity fails** without uniform convergence.

Appendix

Not required for Math Camp. Proofs and extra examples linked from the main slides.

A.1 IFT — several variables

System: $\mathbf{F}(\mathbf{y}, \theta) = \mathbf{0}$, $\mathbf{F}: \mathbb{R}^{n+m} \rightarrow \mathbb{R}^n$.

If the $n \times n$ Jacobian $\frac{\partial \mathbf{F}}{\partial \mathbf{y}}$ is invertible at the point, then locally $\mathbf{y} = \mathbf{y}^*(\theta)$ and

$$\frac{\partial \mathbf{y}^*}{\partial \theta} = - \left(\frac{\partial \mathbf{F}}{\partial \mathbf{y}} \right)^{-1} \frac{\partial \mathbf{F}}{\partial \theta}.$$

Same pattern as the one-variable formulas with F_y and F_θ . Appears in graduate micro; camp focus is the scalar case in section 2.1.

A.2 Proof. EVT on $[a, b]$

for interest only

Claim. If f is continuous on $[a, b]$, then f attains a maximum and minimum on $[a, b]$.

Bolzano–Weierstrass Theorem. Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Step 1. f is bounded on $[a, b]$. Assume, for contradiction, that f is unbounded. Then $\exists (x_n)$ in $[a, b]$ with $|f(x_n)| > n$ for all n . The sequence (x_n) is bounded (it lies in $[a, b]$). By Bolzano–Weierstrass, some subsequence (x_{n_k}) converges to $c \in [a, b]$. Continuity gives $f(x_{n_k}) \rightarrow f(c)$. But $|f(x_{n_k})| > n_k \rightarrow \infty$, a contradiction. Hence f is bounded; let $M = \sup\{f(x) : x \in [a, b]\}$.

A.2 Proof. EVT on $[a, b]$ (continued)

Step 2 — claim. f attains its maximum $M = \sup\{f(x) : x \in [a, b]\}$ on $[a, b]$; i.e. $\exists d \in [a, b]$ with $f(d) = M$.

Why $\exists (y_n)$ in $[a, b]$ with $f(y_n) \rightarrow M$. By definition of supremum:

1. $f(x) \leq M$ for all $x \in [a, b]$;
2. for every $\varepsilon > 0$, $\exists x \in [a, b]$ with $f(x) > M - \varepsilon$.

Take $\varepsilon_n = \frac{1}{n}$: pick $y_n \in [a, b]$ with $f(y_n) > M - \frac{1}{n}$. Since $f(y_n) \leq M$,

$$M - \frac{1}{n} < f(y_n) \leq M \quad \Rightarrow \quad f(y_n) \rightarrow M.$$

A.2 Proof. EVT on $[a, b]$ (continued)

Step 2 — finish.

1. (y_n) is **bounded**: each $y_n \in [a, b] \subset \mathbb{R}$.
2. By Bolzano–Weierstrass, some subsequence (y_{n_k}) converges to a limit $d \in [a, b]$.
3. Since $f(y_n) \rightarrow M$, every subsequence has the **same limit**:

$$f(y_{n_k}) \rightarrow M.$$

4. By **continuity** of f at d ,

$$f(d) = \lim_{k \rightarrow \infty} f(y_{n_k}) = M.$$

So M is attained at $d \in [a, b]$. Minimum: repeat for $-f$.



A.3 Expectation from the CDF

Claim. For non-negative continuous X with PDF f and CDF F ,

$$\mathbb{E}[X] = \int_0^{\infty} xf(x) dx = \int_0^{\infty} (1 - F(x)) dx.$$

Proof (integration by parts). Set $u = 1 - F(x)$ and $dv = dx$. Then $du = -f(x) dx$ and $v = x$:

$$\int_0^{\infty} (1 - F(x)) dx = [x(1 - F(x))]_0^{\infty} - \int_0^{\infty} x(-f(x)) dx = [x(1 - F(x))]_0^{\infty} + \int_0^{\infty} xf(x) dx.$$

At $x = 0$, $x(1 - F(x)) = 0$. As $x \rightarrow \infty$, $x(1 - F(x)) \rightarrow 0$ when $\mathbb{E}[X] < \infty$. Hence

$$\int_0^{\infty} (1 - F(x)) dx = \mathbb{E}[X].$$

□

A.4 Leibniz rule — failure example

Example (fixed limits). On $[0, 1]$, let

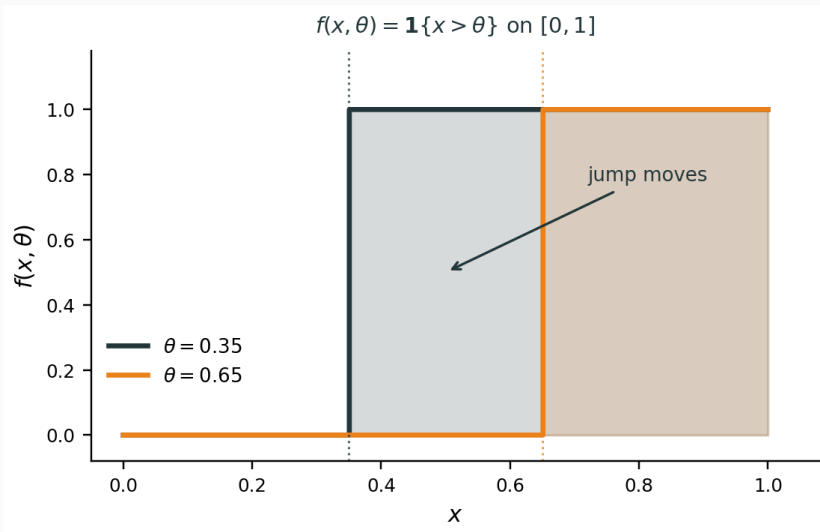
$$f(x, \theta) = \mathbf{1}\{x > \theta\} \quad (\text{step jumps at } x = \theta).$$

Then

$$F(\theta) = \int_0^1 f(x, \theta) dx = 1 - \theta \quad \Rightarrow \quad F'(\theta) = -1.$$

Why Leibniz fails. The discontinuity **moves with** θ , so $\partial f / \partial \theta$ is not continuous on $[0, 1] \times I$. Naively $\partial f / \partial \theta = 0$ a.e., so $\int_0^1 0 dx = 0 \neq F'(\theta)$.

A.4 Leibniz failure — moving step (graph)



A.5 Taylor expansion (one variable)

Claim (second order). If $f \in C^2$ on an interval containing a and $a + h$, then

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2}f''(c)h^2$$

for some c strictly between a and $a + h$.

Proof. Write

$$f(a + h) = f(a) + hf'(a) + R(h).$$

Step 1. By FTC and the chain rule, $f(a + h) - f(a) = \int_0^h f'(a + t) dt$, so

$$R(h) = \int_0^h [f'(a + t) - f'(a)] dt.$$

Step 2. For each $t \in (0, h]$, MVT on f' gives $f'(a + t) - f'(a) = tf''(c_t)$ with $c_t \in (a, a + t)$. Hence

$$R(h) = \int_0^h tf''(c_t) dt = \frac{h^2}{2} f''(c)$$

for some c between a and $a + h$ (continuous f''). □

First order follows from the $n = 1$ case (or MVT on f directly). Multivariable formulas follow from applying the chain rule along $\mathbf{a} + t\mathbf{h}$.

A.6 What is Brownian motion?

Brownian motion W_t (also called a **Wiener process**): a continuous-time **random** path with $W_0 = 0$.

Intuition. Think of a particle jolted by unpredictable shocks (e.g. noise in a stock price). You cannot forecast the next jiggle, but you can describe its **typical size**.

Key facts (informal).

- Over $[t, t + \Delta t]$: increment $\Delta W = W_{t+\Delta t} - W_t$ has mean 0 and variance Δt
- Increments on **non-overlapping** intervals are independent
- Paths are **continuous** but very jagged (not differentiable in the usual sense)

In $dX_t = \mu dt + \sigma dW_t$, the dW_t term is this random noise scaled by σ .

A.6 Taylor and Itô's lemma

Setup. X_t follows an Itô process (e.g. $dX_t = \mu dt + \sigma dW_t$). Over a short interval $[t, t + \Delta t]$,

$$\Delta X = X_{t+\Delta t} - X_t \quad (\text{change in the state variable}).$$

Expand $f(X_t, t)$ in the **two** small steps ΔX and Δt .

Taylor in (x, t) through order 2:

$$\Delta f \approx f_x \Delta X + f_t \Delta t + \frac{1}{2} f_{xx} (\Delta X)^2 + f_{xt} \Delta X \Delta t + \frac{1}{2} f_{tt} (\Delta t)^2.$$

A.6 Itô multiplication rules

Ordinary calculus. If Δt is small, then $(\Delta t)^2$ is **much smaller** and can be dropped.

Stochastic calculus. $\Delta W = W_{t+\Delta t} - W_t$ is one **Brownian increment** (see previous slide): mean 0, variance Δt . So a typical size is $|\Delta W| \sim \sqrt{\Delta t}$, not Δt .

Why $(\Delta W)^2 \approx \Delta t$. Squaring a $\sqrt{\Delta t}$ -sized step gives order Δt :

$$(\Delta W)^2 \sim (\sqrt{\Delta t})^2 = \Delta t.$$

So $(\Delta W)^2$ is **not** negligible next to Δt (unlike $(\Delta t)^2$).

Itô rules (informally): $(\Delta t)^2 \rightarrow 0$, $(\Delta W)^2 \approx \Delta t$, $\Delta t \Delta W \rightarrow 0$. For $dX = \mu dt + \sigma dW$, this makes $\frac{1}{2} f_{xx} (\Delta X)^2$ contribute at the same scale as Δt .

Itô's lemma:

$$df = f_t dt + f_x dX + \frac{1}{2} f_{xx} (dX)^2 \quad (\text{plus higher-order terms}).$$